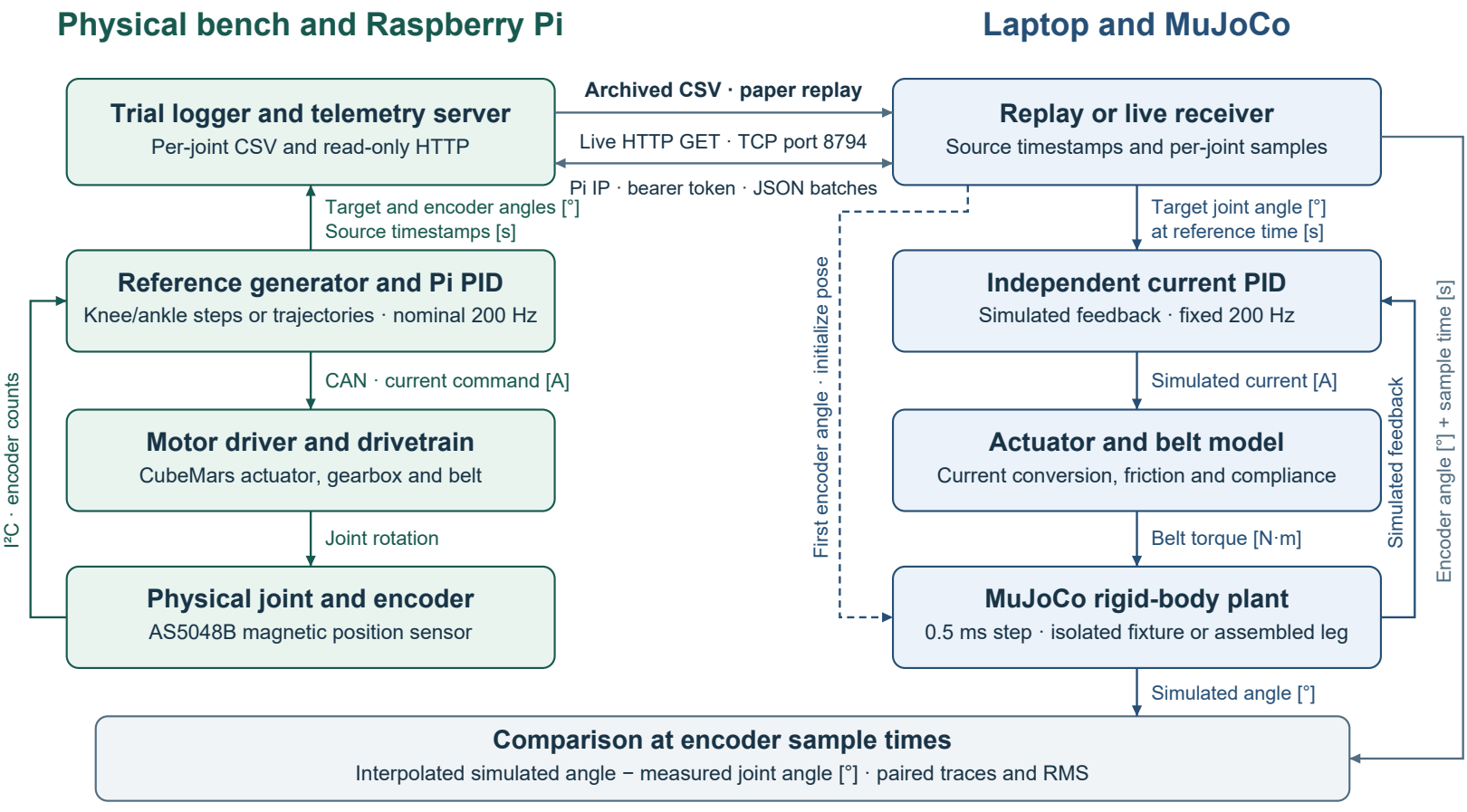

Physical bench and Raspberry Pi
Laptop and MuJoCo
Trial logger and telemetry server
Per-joint CSV and read-only HTTP
Archived CSV · paper replay
Live HTTP GET · TCP port 8794
Pi IP · bearer token · JSON batches
Replay or live receiver
Source timestamps and per-joint samples
Target and encoder angles [°]
Source timestamps [s]
Target joint angle [°]
at reference time [s]
Reference generator and Pi PID
Knee/ankle steps or trajectories · nominal 200 Hz
Independent current PID
Simulated feedback · fixed 200 Hz
I²C · encoder counts
First encoder angle · initialize pose
Simulated feedback
Encoder angle [°] + sample time [s]
CAN · current command [A]
Simulated current [A]
Motor driver and drivetrain
CubeMars actuator, gearbox and belt
Actuator and belt model
Current conversion, friction and compliance
Joint rotation
Belt torque [N·m]
Physical joint and encoder
AS5048B magnetic position sensor
MuJoCo rigid-body plant
0.5 ms step · isolated fixture or assembled leg
Simulated angle [°]
Comparison at encoder sample times
Interpolated simulated angle − measured joint angle [°] · paired traces and RMS